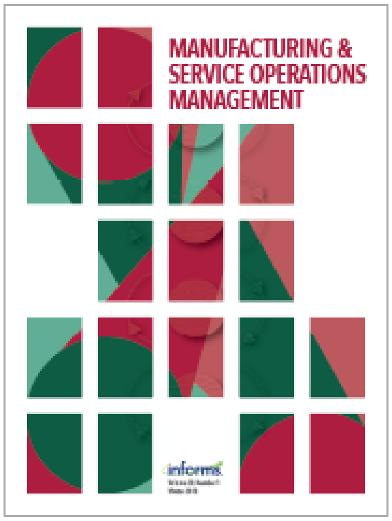


Manufacturing & Service Operations Management

Publication details, including instructions for authors and subscription information:
http://pubsonline.informs.org

Using Mobile Device Data to Understand the Effect of Stay-at-Home Orders on Residents' Mobility

Guihua Wang

To cite this article:
Guihua Wang (2022) Using Mobile Device Data to Understand the Effect of Stay-at-Home Orders on Residents' Mobility. Manufacturing & Service Operations Management 24(6):2882-2900. https://doi.org/10.1287/msom.2021.1014

MANUFACTURING & SERVICE OPERATIONS MANAGEMENT
Vol. 24, No. 6, November–December 2022, pp. 2882–2900
ISSN 1523-4614 (print), ISSN 1526-5498 (online)
https://pubsonline.informs.org/journal/msom

Using Mobile Device Data to Understand the Effect of Stay-at-Home Orders on Residents' Mobility

Guihua Wang[a]

[a] Jindal School of Management, University of Texas at Dallas, Richardson, Texas 75080

Contact: guihua.wang@utdallas.edu, https://orcid.org/0000-0001-8324-7073 (GW)

Received: August 1, 2020
Revised: January 25, 2021; April 13, 2021
Accepted: April 14, 2021
Published Online in Articles in Advance: November 15, 2021

https://doi.org/10.1287/msom.2021.1014

Abstract. ***Problem definition*:** This study addresses three important questions concerning the effectiveness of stay-at-home orders and sociodemographic disparities. (1) What is the average effect of the orders on the percentage of residents staying at home? (2) Is the effect heterogeneous across counties with different percentages of vulnerable populations (defined as those without health insurance or who did not attend high school)? (3) If so, why are the orders less effective for some counties than for others? ***Academic/practical relevance*:** To combat the spread of coronavirus disease 2019 (COVID-19), a number of states in the United States implemented stay-at-home orders that prevent residents from leaving their homes except for essential trips. These orders have drawn heavy criticism from the public because whether they are necessary and effective in increasing the number of residents staying at home is unclear. ***Methodology*:** We estimate the average effect of the orders using a difference-in-differences model, where the control group is the counties that did not implement the orders and the treatment group is the counties that did implement the orders during our study period. We estimate the heterogeneous effects of the orders by interacting county features with treatment dummies in a triple-difference model. ***Results*:** Using a unique set of mobile device data that track residents' mobility, we find that, although some residents already voluntarily stayed at home before the implementation of any order, the stay-at-home orders increased the number of residents staying at home by 2.832 percentage points (or 11.25%). We also find that these orders are less effective for counties with higher percentages of uninsured or less educated (i.e., did not attend high school) residents. To explore the mechanisms behind these results, we analyze the effect of the orders on the average number of work and nonwork trips per person. We find that the orders reduce the number of work trips by 0.053 (or 7.87%) and nonwork trips by 0.183 (or 6.50%). The percentage of uninsured or less educated residents in a county negatively correlates with the reduction in the number of work trips but does not correlate with the reduction in the number of nonwork trips. ***Managerial implications*:** Our results suggest that uninsured and less educated residents are less likely to follow the orders because their jobs prevent them from working from home. Policy makers must take into account the differences in residents' socioeconomic status when developing new policies or allocating limited healthcare resources.

History: This paper has been accepted for the *Manufacturing & Service Operations Management* Special Section on Responsible Research in Operations Management.
Supplemental Material: The online appendix is available at https://doi.org/10.1287/msom.2021.1014.

Keywords: COVID-19 pandemic • sociodemographic disparity • causal inference • mobile device

1. Introduction

The novel coronavirus disease 2019 (COVID-19) is one of the most severe pandemics in recent history. As of November 1, 2020, a total of 47,272,539 confirmed cases and 1,210,225 deaths in the world have been reported. The United States leads the world, with 9,549,011 cases and 236,893 deaths during the same period.[1] COVID-19 is more contagious and dangerous than many other infectious diseases because it can spread through the air over a long distance and remain viable on a variety of surfaces for a long time (Guo et al. 2020, van Doremalen et al. 2020). The disease has an incubation period of 2–14 days, and many patients are infected without knowing it (Li et al. 2020). The combination of high contagiousness and a long incubation period warrants a rapid spread of the disease through person-to-person contacts.

The U.S. federal and local governments have considered various measures (e.g., testing, hospitalization, and quarantine) to contain the spread of the disease. The strictest yet most controversial measures are the stay-at-home orders, which require residents to stay at home except for essential trips. Despite calls from top disease experts for a nationwide lockdown, the federal government has allowed state governments to decide whether to implement the order.[2] On March 19, 2020, California was the first state in the nation to implement the order. Between March and April 2020, a total of 42 states and the District of Columbia implemented statewide stay-at-home orders, and 8 states did not (see Figure 1).[3] Debates regarding whether the orders are necessary and effective in increasing the percentage of residents staying at home have been ongoing.

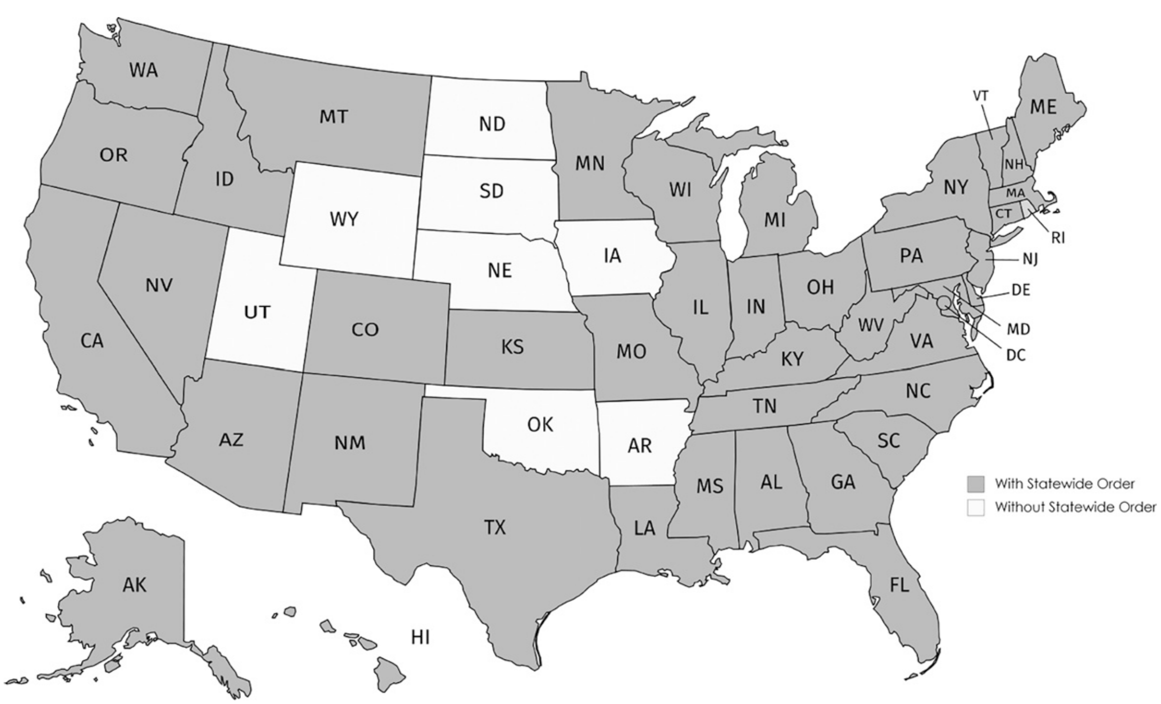
Proponents believe that the orders increase the percentage of residents staying at home for several reasons. First, they raise the awareness about COVID-19. Before the implementation of the orders, despite a growing number of cases reported in the United States, many Americans thought the disease was under control in the country and would disappear in the summer with high temperatures. They were not concerned about the disease, partly because the federal government downplayed the threat of the disease and did not take significant actions beyond instituting international travel restrictions.[4] These orders send a clear message about the serious nature of COVID-19.

Second, these orders prevent residents from leaving their homes except for essential trips such as grocery shopping and prescription refills. Although residents may choose whether to follow voluntary social distancing guidelines without worrying about penalties, they are subject to fines and even jail time when they violate the stay-at-home orders. For example, in Oregon, violating the order is a Class C misdemeanor punishable by up to 30 days in jail, a fine of up to $1,250, or both.[5] Police officers enforce the stay-at-home orders through regular patrols and checkpoints. These financial and criminal penalties may deter residents from leaving their homes to make unnecessary trips.

Third, some of these orders require nonessential businesses (e.g., retail stores, restaurants, and recreational centers) to be closed to the public.[6] The closure of these businesses reduces the number of places residents can visit and therefore, indirectly "forces" residents to stay at home. For example, a closure of retail stores "forces" shoppers to shop online or over the phone, a closure of restaurants "forces" diners to cook for themselves or have food delivered to their houses, and a closure of recreational centers "forces" sports enthusiasts to exercise at home. Some of these orders also encourage companies to adopt work from home policies, which allow employees to reduce the number of work trips to their companies.

Opponents believe these orders do not achieve what they are intended for. First, residents who are health conscious and aware of the risks had already

Figure 1. Status of Statewide Stay-at-Home Order Implementation by State



Notes. In states without statewide orders, some counties and cities implemented their own stay-at-home orders. More details can be found at https://www.nytimes.com/interactive/2020/us/coronavirus-stay-at-home-order.html.

voluntarily stayed at home. Before the implementation of the orders, the World Health Organization (WHO) declared the COVID-19 outbreak a global pandemic,[7] President Trump declared a national emergency,[8] and all states declared state emergencies.[9] Also, the federal and local governments advised residents to practice social distancing by staying at least six feet from other people, staying out of crowded places, and avoiding mass gathering.[10] With these emergency declarations and social distancing guidelines, residents may have already stayed at home, and whether the orders are still necessary is unclear.

Second, residents who do not want to stay at home will still leave their homes even when orders are in place. Staying at home can be challenging for people who are used to gathering in groups and need to care for the elderly or their young children because staying home requires immediate and significant changes to residents' routines and behaviors. Seeing dozens of young people partying in the streets over the weekends, adults walking to hardware stores to buy supplies for their elderly parents' home improvement projects, and parents posting pictures from children's birthday parties and playdates on Facebook is somewhat unsurprising.[11] Staying at home is also challenging for those who see the orders as violations of their constitutional rights and those who worry about economic fallouts. Since the implementation of the orders, an increasing number of residents have taken to the streets and gathered at state capitals to protest the orders.[12]

Third, some residents want to stay at home but do not have the luxury to do so. Residents cannot stay at home if they do not have homes in the first place. An estimated 2 million Americans find themselves homeless in a given year.[13] Among Americans with homes, some cannot stay at home because they work for essential businesses such as gas stations, pharmacies, banks, and laundry services, which remain open during the outbreak of COVID-19. Others cannot stay at home because their jobs require them to be physically present in the companies or with customers. For example, Amazon warehouse order pickers cannot stay at home to pick items ordered by customers, and UPS truck drivers cannot stay at home to deliver packages sent by sellers. These residents cannot stay at home because their jobs prevent them from doing so.

Other than ongoing debates regarding the effectiveness of the orders, concerns about the sociodemographic disparities exacerbated by the pandemic have arisen. Reportedly, in Chicago, Black residents account for 33% of the city's population but 72% of COVID-19–related deaths. In Georgia, White residents account for 58% of the state's population but 40% of COVID-19 cases.[14] At the national level, the Centers for Disease Control and Prevention compiled COVID-19–associated hospitalizations across 99 counties in 14 states in March and found similar results: Black residents account for 18% of the population in these counties but 33% of hospitalizations related to COVID-19.[15]

These alarming disparities have drawn considerable attention from the media, public officials, and researchers. Whereas some speculate that healthcare resources are disproportionately allocated and vulnerable populations do not receive proper healthcare and treatment, others point out that the disparities are because of the differences in genetics and underlying health conditions because some populations, such as Black, uninsured, and less educated, are more likely to have comorbidities, such as diabetes, obesity, and hypertension. Additionally, patients with these comorbidities are more likely to be admitted to hospitals and die from COVID-19.[16]

A widely overlooked factor is the socioeconomic status of vulnerable populations. For example, in Texas, nearly 18% of residents do not have health insurance, and many of those uninsured are unable to take paid sick leave from work.[17] Vulnerable populations such as those uninsured and less educated may be at a higher risk because they have jobs that do not allow them to stay at home. The stay-at-home orders will exacerbate the disparities if they favor only insured and well-educated residents who can work from home.

This study addresses three important questions concerning the effectiveness of stay-at-home orders and sociodemographic disparities. (1) What is the average effect of the orders on the percentage of residents staying at home? (2) Is the effect heterogeneous across different groups of residents, such as those without health insurance or those who did not attend high school? (3) If so, why are the orders less effective for some residents than for others?

The main challenge to addressing the first question is that the orders are not randomly assigned and the states that implemented them may be different from those that did not. We address this challenge using the difference-in-differences approach, where the control group is the counties that did not implement the orders and the treatment group is the counties that did implement the orders during our study period. We include a broad range of county features to capture the systematic differences between the control and treatment groups and day dummies to capture the time trends that affect the percentage of residents staying at home. We also perform extensive robustness checks to analyze the sensitivity of our results.

A major challenge to addressing the second question is that our data are anonymized and aggregated at the county level, so we do not observe individual residents' socioeconomic status and compliance with the stay-at-home orders. This data limitation prevents us from directly comparing two subgroups of residents (e.g., insured versus uninsured residents) for heterogeneous treatment effect analyses. To address this challenge, we leverage the variations across counties in residents' demographics, social, and economic statuses. For example, instead of comparing insured and uninsured residents, we compare counties with different percentages of uninsured residents. More specifically, we first calculate average features of counties in both the control and treatment groups. We then use a difference-in-difference-in-differences (or triple-difference) model to estimate the heterogeneous effects of the orders. As a robustness check, we use a census block group instead of county as a unit of analysis.

Finally, to address the third question and understand why the orders are less effective for some residents, we focus on two types of trips residents make when they leave their homes—work and nonwork trips. We first analyze the average effect of stay-at-home orders on reducing the number of work and nonwork trips using the difference-in-differences model. This approach allows us to understand how residents as a group reduce their work or nonwork trips. We then analyze the heterogeneous effects of the orders on counties with different percentages of vulnerable populations using the triple-difference model. This analysis allows us to understand whether different groups of residents reduce the number of work and nonwork trips by the same amount.

2. Literature Review

There has been a growing interest in using real-word data to analyze the effectiveness of stay-at-home orders in reducing human mobility and containing the COVID-19 pandemic (see Table A1 in Online Appendix A for a list of related studies and their methodologies, data, and study periods). Most of these studies analyzed the average effect of stay-at-home orders and found that the orders are effective in reducing human mobility and the number of COVID-19 cases. It is worth noting that, although the results of these studies are statistically significant, the estimated effects may be biased because of three reasons. First, some of these studies compared outcomes (i.e., human mobility or the number of COVID-19 cases) of the same group before and after the treatment. The results will be biased if there are time trends that affect outcomes even in the absence of the orders. For example, regardless of whether there are orders in place, residents may be more likely to stay at home after seeing an increase in the number of COVID-19 cases. Second, there may be factors that affect both outcomes and decisions to implement the orders. As mentioned in Section 1, a main challenge in analyzing the effect of the stay-at-home orders is that the orders are not randomly assigned and the states that implemented the orders may be different from those that did not. Some of these studies tried to address this challenge using the difference-in-differences model but did not check the critical parallel trends assumption. Finally, some states or counties implemented other social distancing measures (e.g., workplace closures, mask mandates, restaurant restrictions, and school closures) around the same time as the stay-at-home orders. The results will be biased if other measures have significant effects on the outcomes. We contribute to the literature on the average effect of the stay-at-home orders by performing extensive robustness checks to address these issues.

A small but increasing number of studies have examined the heterogeneous effects of the stay-at-home orders. Some of these studies performed state-level analyses using approaches such as the synthetic control method. For example, Guo (2020) studied the effect of the orders on human mobility in the United States and found that the effect of the orders is heterogeneous across different states. Because the results of these studies are aggregated at the state level, it is unclear what types of populations are more likely to follow the orders. A few other studies addressed this challenge using county-level data and interacting county features with the treatment dummy in a regression model. For example, Dave et al. (2021) found that the orders have a larger effect for counties with higher population density and earlier adoption time. Brodeur et al. (2021), Engle et al. (2020), and Grossman et al. (2020) found that the orders have a larger effect for Democrat-leaning counties. Brzezinski et al. (2020c) found that the orders have a smaller effect for counties with a higher concentration of climate change skeptics. Although these studies found a correlation between county features and treatment effect size, they did not explore underlying mechanisms and do not provide empirical evidence why the orders are more effective for some counties than for others.

We make three contributions to the literature on the heterogeneous effects of the stay-at-home orders. First, we focus on vulnerable populations and analyze whether the orders have different effects across counties with different percentages of vulnerable populations. Our study was motivated by news and reports that vulnerable populations are disproportionably affected by COVID-19.[18] Our results are useful to policy makers and researchers in understanding whether the stay-at-home orders exacerbate the disparity. Second, we explore underlying mechanisms of why certain populations are less likely to follow the stay-at-home orders by using a unique set of data from the University of Maryland. There are two types of trips residents make when they leave homes—work and nonwork trips. Existing studies could not separate these two types of trips because their data (e.g., Apple's Mobility Trends Reports, Google Mobility Reports, and SafeGraph's Social Distancing Metrics) do not include information about the number of work and nonwork trips. We bridge this gap in the literature by analyzing the effect of the orders on the number of work and nonwork trips made by different populations. We provide empirical evidence that vulnerable populations are less likely to follow the stay-at-home orders because their jobs do not allow them to work from home. Finally, whereas existing studies analyzed the short-term effect of stay-at-home orders, we include more recent data to examine both the short-term and long-term effects of the orders. We contribute to existing literature by studying how much the long-term effect differs from the short-term effect and whether both long-term and short-term effects are heterogeneous across counties with different percentages of vulnerable populations.

3. Data Description and Preparation

This study builds on three sets of data: stay-at-home orders data, resident mobility data, and county features data. In this section, we first describe each set of the data. We then describe how we prepare the data for empirical analyses.

3.1. Data Description

We obtain stay-at-home-orders data from *The New York Times*, which gathers information from local media coverage and announcements from state and local health agencies.[19] The data include detailed information about the orders, such as implementation dates and order durations. Although some of these orders have different names, such as "shelter in place," "stay-at-home advisory," or "healthy at home," they are essentially the same as stay-at-home orders. Between March and April 2020, a total of 42 states and the District of Columbia implemented statewide stay-at-home orders. The implementation and expiration dates in these states are summarized in Table 1. We see that California and South Carolina are the first and last states, respectively, to implement the orders. Alaska and Georgia are among the first states to lift the orders. Note that some counties and cities implemented stay-at-home orders earlier than their states. For example, Dallas County implemented the order on March 23, 2020, which is earlier than Texas' statewide implementation date of April 2, 2020. In these cases, we use the implementation dates of the counties or cities instead of the states because the former better captures the effect of the orders.

We obtain resident mobility data from The University of Maryland COVID-19 Impact Analysis Platform, which reports residents' mobility based on the data

Table 1. Orders Implementation and Expiration Dates by State

State	Implementation	Expiration	State	Implementation	Expiration
Alabama	04/04/20	04/30/20	Missouri	04/06/20	05/04/20
Alaska	03/28/20	04/24/20	Montana	03/28/20	04/26/20
Arizona	03/31/20	05/15/20	Nebraska		
Arkansas			Nevada	03/31/20	05/09/20
California	03/19/20		New Hampshire	03/28/20	06/15/20
Colorado	03/26/20	04/26/20	New Jersey	03/21/20	06/09/20
Connecticut	03/23/20	05/20/20	New Mexico	03/24/20	05/31/20
Delaware	03/24/20	05/31/20	New York	03/22/20	06/13/20
District of Columbia	04/01/20	05/29/20	North Carolina	03/30/20	05/22/20
Florida	04/03/20	04/30/20	North Dakota		
Georgia	04/03/20	04/24/20	Ohio	03/24/20	05/29/20
Hawaii	03/25/20	05/31/20	Oklahoma		
Idaho	03/25/20	04/30/20	Oregon	03/23/20	05/15/20
Illinois	03/21/20	05/31/20	Pennsylvania	04/01/20	06/04/20
Indiana	03/25/20	05/01/20	South Carolina	04/07/20	05/04/20
Iowa			South Dakota		
Kansas	03/30/20	05/03/20	Tennessee	04/01/20	04/30/20
Kentucky	03/26/20	05/11/20	Texas	04/02/20	04/30/20
Louisiana	03/23/20	05/15/20	Utah		
Maine	04/01/20	05/31/20	Vermont	03/24/20	05/15/20
Maryland	03/30/20	05/15/20	Virginia	03/30/20	06/10/20
Massachusetts	03/24/20	05/18/20	Washington	03/23/20	05/31/20
Michigan	03/24/20	05/28/20	West Virginia	03/24/20	05/04/20
Minnesota	03/28/20	05/17/20	Wisconsin	03/25/20	05/13/20
Mississippi	04/03/20	04/27/20	Wyoming		

Notes. In states with statewide orders, some counties/cities implemented stay-at-home orders earlier than their states. In states without statewide orders, some counties/cities implemented their own stay-at-home orders. More details can be found at https://www.nytimes.com/interactive/2020/us/coronavirus-stay-at-home-order.html.

from mobile devices, government agencies, and healthcare systems.[20] One nice feature of the data is that the results are aggregated at the county level over different days, which allows us to compare the mobility of residents in different counties before and after the implementation of stay-at-home orders. Another nice feature is that the data include different categories of mobility, such as the percentage of residents staying at home and the average number of trips per person, where staying at home is defined as not moving more than one mile away from home based on mobile devices. In addition, the data differentiate between work (e.g., schools and companies) and nonwork (e.g., retails and restaurants) trips based on the origin or destination of the trips, which allows us to understand whether residents reduce their work or nonwork trips to stay at home.

We obtain county features data from the American Community Survey (ACS), which gathers county-level information about residents' demographics and social economic status.[21] We select features that are relevant to this study based on two criteria: (1) the features help predict the percentage of residents staying at home and (2) the features do not create a multicollinearity issue. Our final set of features includes residents' age, race, ethnicity, education attainment, insurance status, household income, population size, land area, and average commute time to work. The ACS data report the values of these features based on one- and five-year estimates. The one-year estimates are suitable for understanding short-term changes in medium to large counties because the data include only counties with a population of at least 65,000 people. The five-year estimates are more suitable for us because the data include counties of various population sizes.

3.2. Data Preparation

We merge resident mobility data with county features data using the names of the counties. After excluding 36 counties that cannot be merged, we obtain a total of 3,106 counties in 50 states and the District of Columbia. In some cases, different cities in the same county implemented stay-at-home orders on different dates. For example, in Oklahoma County, Oklahoma City implemented the order on March 28, 2020, whereas Edmond implemented the order on March 30, 2020.[22] As another example, in DeKalb County, Brookhaven, Avondale Estates, and Doraville implemented their own stay-at-home orders before the county implemented a countywide order on March 28, 2020. We exclude 11 such counties with multiple implementation dates because our unit of analysis is a county. Our final sample includes 3,095 counties. Finally, given the availability of the mobility data, we focus on the period between January 1, 2020, and October 31, 2020. All counties in the treatment group have more than two months in both pre- and post-treatment periods, which allows us to compare residents' mobility before and after the implementation of orders.

4. Econometric Model

In this section, we first discuss our empirical strategy for estimating the effect of stay-at-home orders. We then describe our variables of interest and the difference-in-differences model for average treatment effect analysis. Finally, we describe the triple-difference model for heterogeneous treatment effect analysis.

4.1. Empirical Strategy

To analyze the effect of stay-at-home orders, one approach is to compare percentages of residents staying at home in the states that implemented the orders with those that did not to see how the outcomes are different between these two groups of states. This approach may lead to a biased estimate of the treatment effect if there exist factors that affect both the percentage of residents staying at home and the decision to implement the orders. For example, states with higher population densities may be more likely to implement the orders. At the same time, these states may have higher percentages of residents staying at home.

Another approach is to focus on the states that implemented the orders and compare percentages of residents staying at home before and after the implementation of the orders. This approach will lead to a biased estimate of the treatment effect if there exist time trends that affect the percentage of residents staying at home even in the absence of the orders. For example, residents may be less likely to stay at home in the spring than in the summer because of the difference in weather. As another example, residents may be more likely to stay at home after WHO's declaration of COVID-19 as a pandemic and governments' emergency declarations.

We address these challenges by first defining the control group as the counties that did not implement the stay-at-home orders and the treatment group as the counties that implemented the orders during our study period. We then compare the outcome trends of these two groups before and after the treatment. A change of trend after the treatment would suggest that the orders change the percentage of residents staying at home. Finally, we use a difference-in-differences model to estimate the effect of the orders. This model captures the systematic differences between the control and treatment groups, as well as the time trends that affect the percentage of residents staying at home.

To analyze the heterogeneous effects of the orders, an ideal approach is to stratify residents into different groups and then perform separate analyses for each group. Unfortunately, our data are aggregated at the county level, and we do not observe individual residents' features or outcomes. Our empirical strategy relies on comparing counties with different mixes of residents. For example, instead of comparing insured and uninsured residents, we compare counties with different percentages of uninsured residents. More specifically, we estimate heterogeneous effects using a triple-difference model, where we interact the county feature of interest with the dummies of the treatment, states, and days.

4.2. Average Effect Analysis

To describe the model for average effect analysis, we let $StayAtHome_{it}$ denote the percentage of residents staying at home in county i on day t. In the main analysis, we use $StayAtHome_{it}$ as the dependent variable. As a robustness check, we use $\ln(StayAtHome_{it})$ as an alternative dependent variable. Our independent variable of primary interest is a treatment dummy, $OrderImplementation_{it}$, that equals one if (1) county i implemented a stay-at-home order and (2) the order was implemented on day t or earlier. Similarly, we denote by $OrderExpiration_{it}$ a dummy that equals one if (1) county i implemented a stay-at-home order and (2) the order expired on day t or earlier.

Two approaches can be taken to control for the differences across counties. One approach is to include county fixed effects (i.e., $County_i$), and the other approach is to include county features (i.e., $Feature_i$). We choose the latter approach for two reasons. First, we are interested in understanding what types of counties have higher percentages of residents staying home. Second, the latter approach is more suitable for heterogeneous effect analysis because we need to interact the feature of interest with the dummies of the treatment, states, and days. As a robustness check, we use county fixed effects to estimate the average treatment effect.

The features we consider in this study include residents' demographics (e.g., age, race, and ethnicity) and socioeconomic statuses (e.g., education, insurance status, and household income) aggregated at the county level. We also include counties' population sizes, land areas, and average commute time to work because they may affect the percentage of residents staying at home.[23] Note we do not include time-variant features because changes in these features may be because of the treatment. As another robustness check, we include a lagged version of the dependent variable as an additional feature to reestimate the treatment effect.

To control for the systematic differences across these counties' states, we include a vector of state dummies (denoted by $State_{j(i)}$), which equal to one if county i is in state j and zero otherwise. We also include a vector of day dummies (denoted by Day_t) to control for time trends of residents staying at home because residents may be more likely to stay at home on some days (e.g., weekends) than others (e.g., weekdays). To fully control for time fixed effects, we let the dimension of the vector equal the number of days in this study instead of the number of days in a week.

We estimate the effect of the orders on the percentage of residents staying at home using a difference-in-differences model (Angrist and Pischke 2008):

$$\begin{aligned} StayAtHome_{it} = {} & \alpha_0 + \alpha_1 OrderImplementation_{it} \\ & + \alpha_2 OrderExpiration_{it} x + \alpha_3 State_{j(i)} \\ & + \alpha_4 Day_t + \alpha_5 Feature_i + \varepsilon_{it}, \end{aligned} \tag{1}$$

where ε_{it} denotes an idiosyncratic error.

We are particularly interested in the coefficient α_1 because it indicates the average effect of the orders on the percentage of residents staying at home. A positive coefficient would suggest the orders increase the percentage of residents staying at home, whereas a negative coefficient would suggest otherwise. Similarly, the coefficient α_2 indicates the effect of the expiration of the orders on the percentage of residents staying at home. The coefficients α_3, α_4, and α_5 indicate which states and days and what types of counties, respectively, have higher percentages of residents staying at home.

4.3. Heterogeneous Effect Analysis

The model for heterogeneous effect analysis builds on the difference-in-differences model described in Section 4.2. Our objective is to analyze whether the effect of the orders differs across counties with different features. Let $Target_i$ denote a target feature of interest (e.g., percentage of residents who are uninsured).[24] We estimate the heterogeneous effects of the orders across the target feature using the triple-difference model:

$$\begin{aligned} StayAtHome_{it} = {} & \alpha_0 + \alpha_1 OrderImplementation_{it} \\ & + \beta_1 OrderImplementation_{it} \times Target_i \\ & + \alpha_2 OrderExpiration_{it} \\ & + \beta_2 OrderExpiration_{it} \times Target_i \\ & + \alpha_3 State_{j(i)} + \beta_3 State_{j(i)} \times Target_i \\ & + \alpha_4 Day_t + \beta_4 Day_t \times Target_i \\ & + \alpha_5 Feature_i + \varepsilon_{it}, \end{aligned} \tag{2}$$

where $OrderImplementation_{it} \times Target_i$, $OrderExpiration_{it} \times Target_i$, $State_{j(i)} \times Target_i$, and $Day_t \times Target_i$ denote the interactions between the target feature and dummies of order implementation, order expiration, states, and days, respectively, and ε_{it} denotes an idiosyncratic error.

The coefficient of primary interest is now β_1, which indicates the relative effect of the orders across feature *i*. A coefficient that is significantly different from zero would suggest the effect of the orders is heterogeneous across counties with different values of $Target_i$, whereas an insignificant coefficient would suggest the effect is homogeneous.

We consider two different models in which $Target_i$ refers to (1) percentage of residents who are uninsured and (2) percentage of residents who did not attend high school. We are interested in understanding whether the orders have heterogeneous effects across counties with different percentages of these vulnerable populations.

5. Average Effect and Robustness

In this section, we first provide descriptive results from the summary statistics of the dependent variable and county features and check the parallel trends assumption. We then describe the results from average treatment effect analysis. Finally, we discuss robustness checks to analyze the sensitivity of our results.

5.1. Descriptive Results

Our study includes 509 counties in the control group and 2,586 counties in the treatment group. Table 2 provides definitions and summary statistics of the dependent variable and county features. We calculate the mean and standard deviation of a variable using the values of individual counties.[25] Comparing the control and treatment groups, we see the percentage of residents staying at home is 21.10 in the control group and 21.51 in the treatment group, which suggests the treatment group has a higher percentage of residents staying at home.

The control and treatment groups are similar across some dimensions of features. For example, the median age is around 41, and average household income is around $65,000 in both groups. These two groups are different across other dimensions. For example, the control group has a lower percentage of residents who are Black and a smaller population size than the treatment group. These comparisons suggest that the control and treatment groups are not the same.[26] Fortunately, the difference-in-differences model does not require the control and treatment groups to be the same along all dimensions of features. It requires that the two groups have parallel trends of the dependent variable in the absence of treatment.

To check the parallel trends assumption, we compare the dependent variables of the control and treatment groups over different days. Figure 2 illustrates the comparison, where the two vertical dashed lines indicate the start dates of the first (March 19, 2020) and last (April 7, 2020) orders implemented by different states. From January 1, 2020 to March 19, 2020 (i.e., before the implementation of any order), the two groups have almost the same percentage of residents staying at home.

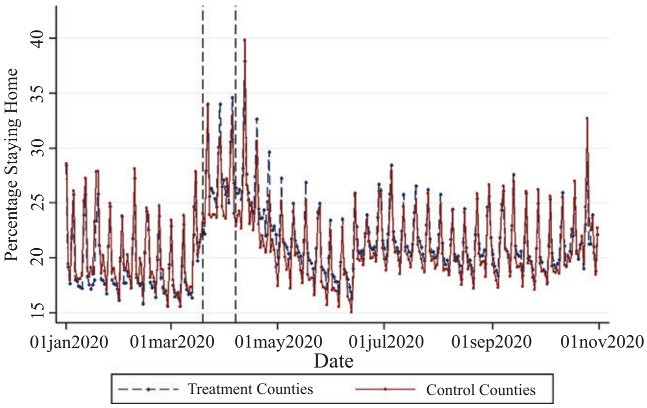
In particular, we see that both groups have a higher percentage of residents staying at home over weekends than weekdays. Also, both groups have an increase in the percentage of residents staying at home

Table 2. Definition and Summary of the Dependent Variable and County Features

		Control group		Treatment group	
Variable	Definition	Mean	Standard deviation	Mean	Standard deviation
Dependent variable					
StayAtHome	Percentage of residents staying at home	21.10	7.30	21.51	6.66
County features					
AgeMedian	Median age in years	41.26	5.66	41.20	5.29
RaceWhite	Percentage of residents who are White	87.10	15.22	82.81	16.65
RaceBlack	Percentage of residents who are Black	3.51	8.85	10.05	15.08
RaceOthers	Percentage of residents who are of other races	9.39	13.04	7.14	8.65
HispanicYes	Percentage of residents who are Hispanic	5.53	5.88	9.76	14.58
HispanicNo	Percentage of residents who are non-Hispanic	94.47	5.88	90.24	14.58
HighSchoolYes	Percentage of residents who attended high school	89.00	4.88	85.61	6.61
HighSchoolNo	Percentage of residents who did not attend high school	11.00	4.88	14.39	6.61
InsuranceYes	Percentage of residents who are insured	89.73	5.37	88.77	5.03
InsuranceNo	Percentage of residents who are uninsured	10.27	5.37	11.23	5.03
CommuteTime	Average commuting time (in minutes) to work	19.66	4.23	24.16	5.50
HouseholdIncome	Average household income in thousands of U.S. dollars	65.79	12.08	64.96	16.60
Population	Number of residents (in thousands) in a county	27.81	56.60	113.19	351.06
LandArea	Land area in thousands of square miles	1.12	1.18	1.03	3.38
Number of counties		509		2,586	

Notes. Staying at home means that no trips are more than one mile away from home. The mean is an unweighted average of county outcomes/features.

Figure 2. (Color online) Compare Control and Treatment Counties (Percentage of Residents Staying at Home)



Notes. This figure compares the percentage of residents staying at home in the control and treatment groups over different days. The two vertical dashed lines indicate the first (March 19, 2020) and last (April 7, 2020) orders implemented in different states.

around March 11, 2020, the day when the World Health Organization declared COVID-19 a pandemic. Finally, both groups have an increase in the percentage of residents staying at home before the implementation of the stay-at-home orders. It is possible that the COVID-19 information kept some risk-averse residents staying at home prior to the orders. This COVID-19 information effect does not bias our results from the difference-in-differences approach because both the control and treatment groups behave in the same manner.

From March 19, 2020 (i.e., after the implementation of some orders), the two groups have different percentages of residents staying at home, as indicated by the gap between the two curves. More specifically, we see that the treatment group has a relative increase in the percentage of residents staying at home after the treatment. This relative increase suggests the stay-at-home orders increase the percentage of residents staying at home. Interestingly, the control and treatment groups have similar trends after the treatment as well, suggesting a sharp effect of the stay-at-home orders.

5.2. Average Treatment Effect

We formally analyze the effect of the orders using the difference-in-differences model (i.e., Equation (1)). Table 3 summarizes the results. The total number of observations is (509 control counties + 2,586 treatment counties) × 305 days = 943,975. From the table, we see that the coefficient of *OrderImplementation* is positive and significantly different from zero at the 1% significance level. This result suggests that the orders increase the percentage of residents staying at home. More specifically, a coefficient of 2.832 means the orders increase the number of residents staying at home by 2.832 percentage points (or 11.25%).

Table 3. Results from the Difference-in-Differences Model

Variable	Coefficient	Standard error
OrderImplementation	2.832***	0.326
OrderExpiration	−1.484***	0.264
AgeMedian	0.037	0.034
RaceBlack	0.032**	0.013
RaceOthers	0.061**	0.026
HispanicYes	0.062***	0.016
HighSchoolYes	0.088**	0.041
InsuranceYes	0.048	0.053
CommuteTime	−0.253***	0.030
HouseholdIncome	0.050***	0.014
Population	0.002**	0.001
LandArea	0.019	0.023
DayFixedEffects	Included	
StateFixedEffects	Included	
Number of observations	943,975	
Adjusted R^2	0.443	

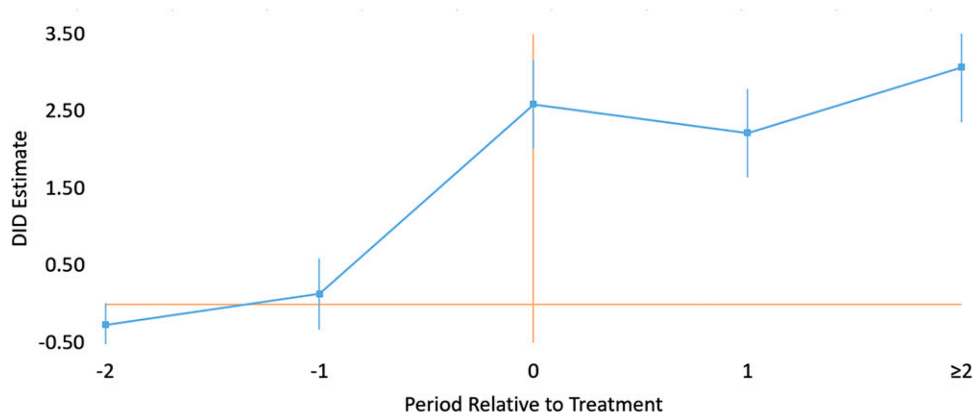
Notes. This table summarizes the results from the difference-in-differences model, where the dependent variable is the percentage of residents staying at home and independent variables are the implementation and expiration of stay-at-home orders, county features, and day and state fixed effects. Robust standard errors are clustered by state.
$^{**}p < 0.05$; $^{***}p < 0.01$.

The coefficient of *OrderExpiration* is negative and significantly different from zero at the 1% significance level, which suggests that the expiration of the orders reduces the percentage of residents staying at home. More specifically, a coefficient of −1.484 means the expiration of the orders reduces the percentage of residents staying at home by 1.484 percentage points (or 5.46%). Our estimation shows that the coefficients of *OrderImplementation* and *OrderExpiration* have opposite signs, which presents more compelling evidence that the orders increase the percentage of residents staying at home. The coefficient of *OrderExpiration* has a smaller magnitude than that of *OrderImplementation*, which suggests that the percentage of residents staying at home has not reached preimplementation level.

From the table, we also see that counties that have higher percentages of residents staying at home are those with higher percentages of residents who are non-White, are Hispanic, and attended high school. Counties that have lower percentages of residents staying at home are those with longer commute times to work, lower household incomes, and smaller population sizes. We note that these coefficients indicate the correlational instead of causal relationship between these features and the dependent variable.

The average effect analysis we have discussed so far does not indicate when the changes happen. Residents may have started staying at home even before the implementation of the orders because the orders are usually announced several days before the implementation dates. Our analysis does not indicate

Figure 3. (Color online) Average Effect of the Orders on Staying at Home (with Pre- and Posttreatment Dummies)



Notes. This figure provides a visual illustration of the results from the difference-in-differences (DID) model with pre- and posttreatment dummies. The DID estimates are based on the coefficients of *OrderImplementation* with a 90% confidence interval. One period corresponds to three days. The omitted (i.e., control) period is three or more periods before the treatment; "≥ 2" indicates two or more periods after the treatment.

whether the effect of the orders increases, decreases, or remains the same over time. Residents may be less likely to follow the orders when they feel bored staying at home or more likely to stay at home when they see others doing so.

To address the first question, we include two pretreatment dummies, $OrderImplementation_{t-1}$ and $OrderImplementation_{t-2}$, to indicate whether the outcome is measured one period (i.e., 1–3 days) or two periods (i.e., 4–6 days) before the implementation dates. Note that the omitted (i.e., control) period is three or more periods before the implementation dates. To address the second question, we include a dummy, $OrderImplementation_t$, to indicate the starting period of the orders and two dummies, $OrderImplementation_{t+1}$ and $OrderImplementation_{t+2}$, to indicate whether the outcome is measured one period or two (or more) periods after the starting period of the orders.

The results from this modified difference-in-differences model are summarized in Figure 3 and Table 4. The coefficients of the pretreatment dummies have small magnitudes and are not significantly different from zero, suggesting that the control and treatment groups are not significantly different from each other in pretreatment periods. These results further support the parallel trends assumption made in the difference-in-differences model. From the coefficients of the treatment and posttreatment dummies, we see that the effect of the orders remains almost the same over time. The significance of these posttreatment dummies also indicates that the insignificance of the pretreatment dummies is not because of a small sample size.

5.3. Robustness Checks

We conduct nine robustness checks to analyze the sensitivity of our results. The first two robustness checks address potential concerns that our results are driven by a particular state or period of time. The third robustness check addresses a potential concern that our results are confounded by other social distancing measures. The fourth robustness check uses county fixed effects instead of county features to control for the systematic differences across counties. The fifth robustness check addresses potential endogeneity issues. The six robustness check uses an alternative treatment group. The seventh robustness check includes a lagged version of the dependent variable as an additional independent variable. The eighth and ninth robustness checks use an alternative dependent variable and data set, respectively. In all robustness checks, we cluster robust standard errors by state. Detailed results from these robustness checks are summarized in Online Appendix C.

Table 4. Average Effect of the Orders on Staying at Home (with Pre- and Posttreatment Dummies)

Variable	Coefficient	Standard error
OrderImplementation		
More than two periods before	Control	
Two periods before	−0.267	0.172
One period before	0.133	0.281
Period of treatment	2.591***	0.350
One period after	2.220***	0.432
Two or more periods after	3.070***	0.392
OrderExpiration	Included	
CountyFeatures	Included	
DayFixedEffects	Included	
StateFixedEffects	Included	
Number of observations	943,975	
Adjusted R^2	0.443	

Notes. This table summarizes the results from the difference-in-differences model with pre- and posttreatment dummies. One period corresponds to three days. The omitted (i.e., control) period is three or more periods before the treatment. Robust standard errors are clustered by state.

***$p < 0.01$.

5.3.1. Leave-State-Out Analysis. This robustness check addresses a potential concern that our results are driven by a particular state. The orders may have a large effect on one state but a small or negligible effect on the other states. We address this concern by leaving one state out each time and reestimating the treatment effect. Table C1 in Online Appendix C summarizes the results from this leave-state-out analysis. We see that all coefficients are positive and significantly different from zero. These coefficients are not significantly different from each other and that in the main analysis, suggesting that our result is not driven by a particular state.

5.3.2. Leave-Days-Out Analysis. Another potential concern is that our results are driven by a particular period of time. For example, the orders may have a large effect on some days but a small or negligible effect on the other days. We address this concern by performing three different leave-days-out analyses in which we leave out (1) the same day (e.g., Mondays) of a week, (2) weekdays or weekends, and (3) a specific month. Table C2 in Online Appendix C summarizes the results from these leave-days-out analyses. We see that all coefficients are positive and significantly different from zero. These coefficients are not significantly different from each other and that in the main analysis, suggesting that our result is not driven by a particular period of time.

5.3.3. Other Statewide Policies. This robustness check addresses a potential concern that some states implemented other social distancing measures during our study period. We cannot attribute the observed effects to stay-at-home orders if a state implemented the stay-at-home orders and other measures at the same time. Different states implemented mainly four other social distancing measures during our study period—workplace closures, mask mandates, restaurant restrictions, and school closures. We obtain the implementation dates of these measures in different states from the Boston University COVID-19 U.S. State Policy Database.[27]

The first four rows of Table C3 in Online Appendix C summarize the results when we control for workplace closures, mask mandates, restaurant restrictions, and school closures, respectively. The fifth row of Table C3 in Online Appendix C summarizes the result when we control for all measures at the same time. We see that all coefficients are positive and significantly different from zero. These coefficients are not significantly different from each other and that in the main analysis, suggesting that our result is not confounded by other measures.

5.3.4. County Fixed Effects. In the main analysis, we use county features to control for the differences across counties. A potential concern is that our model does not include all features that affect both the percentage of residents staying at home and the decision to implement the stay-at-home orders. We address this concern by using county fixed effects instead of county features in the difference-in-differences model. Table C3 (row "County fixed effects") in Online Appendix C summarizes the results from this robustness check. We see that the coefficient has almost the same magnitude as that in the main analysis, suggesting that our result is not driven by the use of county features.

5.3.5. Potential Endogeneity Issues. Although the county fixed effects model addresses potential selection issues related to time-invariant unobservable features, there may exist time-variant unobservable features that affect both the percentage of residents staying at home and the decision to implement the stay-at-home orders. To address this concern, we follow the strategy in the extant literature (see, e.g., Brot-Goldberg et al. 2017, Bapna et al. 2018, Sun et al. 2020) by using counties with early implementation dates as a control group and those with late implementation dates as a treatment group. This approach could capture some of the time-variant unobservable features because both the control and treatment counties eventually implemented the orders.

Table C3 (row "Potential endogeneity issues") in Online Appendix C summarizes the results when we use counties in the states (California, Connecticut, Illinois, Louisiana, New Jersey, New York, Ohio, Oregon, and Washington) that implemented the orders between March 19, 2020 and March 23, 2020 as a control group and counties in the states (Alabama, District of Columbia, Florida, Georgia, Maine, Missouri, Mississippi, Nevada, Pennsylvania, South Carolina, and Texas) that implemented the orders between April 1, 2020 and April 7, 2020 as a treatment group.[28] From the table, we see that the coefficient of *Order-Implementation* is positive and significantly different from zero, which suggests that endogeneity issues are not a major concern in this study.

5.3.6. Alternative Treatment Group. In the main analysis, our control group includes 8 states in the Midwest or nearby regions, and the treatment group includes 43 states that spread widely across the United States. A natural question is whether we could further improve our analysis by focusing on states that are geographically close to each other. In this robustness check, we use states that are neighbors to the control group as an alternative treatment group. Table C3 (row "Alternative treatment group") in Online Appendix C summarizes the results when the treatment group includes counties in Colorado, Illinois, Kansas, Minnesota, Montana, and Wisconsin. We see that the coefficient of *OrderImplementation* is positive and significantly different from zero, suggesting that our result is robust to an alternative treatment group specification.

5.3.7. Additional Independent Variables. This robustness check addresses a potential concern that outcomes measured on the same day of a week are positively correlated. For example, the percentage of residents staying at home on Saturdays may be correlated because of repeating patterns of routine. We address this concern using two different models in which we include (1) the baseline percentage of residents staying at home, for the corresponding day of the week, between February 23, 2020 and February 29, 2020[29] and (2) a lagged version of the dependent variable as an additional independent variable.

The results from these two models are summarized in Table C3 (rows "Baseline StayAtHome" and "Lagged StayAtHome," respectively) in Online Appendix C. We see all coefficients are positive and significantly different from zero. Note that the magnitude of the coefficients in row "Lagged StayAtHome" is smaller than that in the main analysis because the orders increase the percentage of residents staying at home. The positive correlation between the orders and a lagged version of the dependent variable leads to an underestimation of the treatment effect.

5.3.8. Alternative Dependent Variable. In this robustness check, we use the log-transformed percentage of residents staying at home, ln*(StayAtHome)*, as an alternative dependent variable and reestimate the treatment effect. The result is summarized in Table C3 (row "ln(StayAtHome)") in Online Appendix C. We see that the coefficient of *OrderImplementation* is positive and significantly different from zero. More specifically, a coefficient of 0.113 means that the order increases the percentage of residents staying at home by 11.3%. Note that the adjusted R^2 in this robustness check is smaller than that in the main analysis because the original variable follows a normal distribution and better fits the difference-in-differences model.

5.3.9. Alternative Data. To address a potential concern that our results are driven by particular data, we use alternative data from SafeGraph to check the robustness of our results. SafeGraph is a location and business intelligence firm that obtains global positioning system data by regularly pinging 18 million smartphones with certain apps each day.[30] We focus on SafeGraph's Social Distancing Metrics data because they record the percentage of devices that are completely at home. A nice feature of the data is that the devices are aggregated at the census block group level, which allows us to observe residents' mobility at a more granular level.[31] Table C3 (row "Alternative data") in Online Appendix C summarizes the result from this robustness check. We see that the coefficient of *OrderImplementation* is positive and significantly different from zero, which suggests that stay-at-home orders increase the percentage of residents staying at home. Interestingly, the result from this robustness check is similar to that in the main analysis, which suggests that our result in the main analysis is not driven by particular data.

6. Heterogeneous Effects and Discussion

In this section, we first describe the results from heterogeneous treatment effect analysis. We then explore mechanisms behind the results by analyzing the effect of the orders on the number of work and nonwork trips.

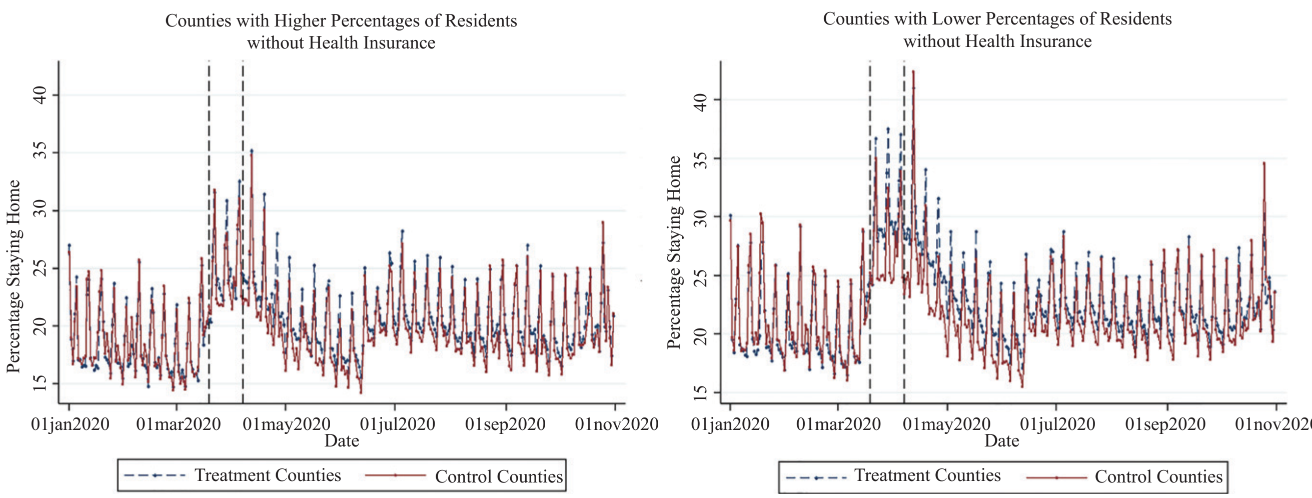
6.1. Heterogeneous Treatment Effects

We focus on two target features: (1) percentage of residents without health insurance and (2) percentage of residents who did not attend high school. For each target feature, we first divide all counties into two categories based on the median value of the feature. We then examine the treatment effect heterogeneity by comparing the outcomes of these two categories. Then, we estimate heterogeneous treatment effects using the triple-difference model (i.e., Equation (2)).

6.1.1. Heterogeneity Across Insurance. To examine the treatment effect heterogeneity across insurance, we divide counties into two categories of equal size based on the percentage of residents without health insurance. We informally designate these two categories as "Counties with Higher Percentages of Residents without Health Insurance" and "Counties with Lower Percentages of Residents without Health Insurance." For each category, we compare the percentage of residents staying at home in control and treatment groups (see Figure 4). In the first category with higher percentages of residents without health insurance, the control and treatment groups have similar trends both before and after the treatment. In the second category with lower percentages of residents without health insurance, the two groups have almost the same trend before the treatment and remarkably different trends after the treatment. The comparisons suggest that the orders are less effectiveness for counties with higher uninsured rates.

We formally analyze the treatment effect heterogeneity across insurance by interacting percentage of uninsured residents with the dummies of the treatment, days, and states in the triple-difference model. Note that the percentage of uninsured residents is a

Figure 4. (Color online) Comparison by Percentage of Residents Without Health Insurance



Notes. We divide counties into two categories of equal size based on the percentage of uninsured residents. We informally designate these two categories as "Counties with Higher Percentages of Residents without Health Insurance" and "Counties with Lower Percentages of Residents without Health Insurance." The two vertical dashed lines indicate the first (March 19, 2020) and last (April 7, 2020) orders implemented in different states.

continuous feature in this analysis. The second column of Table 5 summarizes the results. We see that the coefficient of *OrderImplementation* × *Uninsured* is negative and significantly different from zero. This result suggests that the orders have a smaller effect on counties with higher percentages of uninsured residents. More specifically, a coefficient of −0.108 means that a 1-percentage point increase in the number of uninsured residents reduces the effect of the orders by 0.108 percentage points (or 0.30%).

The coefficient of *OrderExpiration* × *Uninsured* is positive and significantly different from zero, suggesting that the expiration of the orders has a smaller effect on counties with higher percentages of uninsured residents. More specifically, a coefficient of 0.171 means that a 1-percentage point increase in the number of uninsured residents reduces the effect of the expiration of the orders by 0.171 percentage points (or 0.78%). Our estimation shows that the coefficients of *OrderImplementation* × *Uninsured* and *OrderExpiration* × *Uninsured* have opposite signs, which presents more compelling evidence that the orders have a smaller effect on counties with higher percentages of uninsured residents.

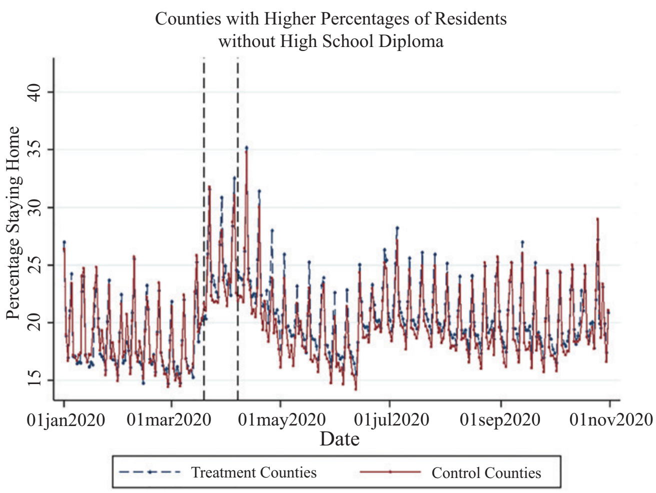
6.1.2. Heterogeneity Across Education. Similarly, to examine the treatment effect heterogeneity across education, we divide counties into two categories of equal size based on the percentage of residents who did not attend high school. We informally designate these two categories as "Counties with Higher Percentages of Residents without High School Diploma" and "Counties with Lower Percentages of Residents

Table 5. Heterogeneous Effects of the Orders on Staying at Home

	(1) *Uninsured*		(2) *NoHighSchool*	
Variable	Coefficient	Standard error	Coefficient	Standard error
OrderImplementation	3.692***	0.542	3.630***	0.650
× *TargetFeature*	−0.108**	0.038	−0.069*	0.042
OrderExpiration	−3.113***	0.543	−2.512***	0.457
× *TargetFeature*	0.171***	0.045	0.071**	0.030
CountyFeatures	Included		Included	
DayFixedEffects	Included		Included	
× *TargetFeature*	Included		Included	
StateFixedEffects	Included		Included	
× *TargetFeature*	Included		Included	
Number of observations	943,975		943,975	
Adjusted R^2	0.464		0.466	

Notes. In columns (1) and (2), the target features are *Uninsured* and *NoHighSchool*, respectively. Robust standard errors are clustered by state.
*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

Figure 5. (Color online) Comparison by Percentage of Residents Without High School Diploma



Notes. We divide counties into two categories of equal size based on the percentage of residents who did not attend high school. We informally designate these two categories as "Counties with Higher Percentages of Residents without High School Diploma" and "Counties with Lower Percentages of Residents without High School Diploma." The two vertical dashed lines indicate the first (March 19, 2020) and last (April 7, 2020) orders implemented in different states.

without High School Diploma." Figure 5 compares the percentages of residents staying at home for each category. We see that the control and treatment groups in both categories have almost the same trends before the treatment. After the treatment, the treatment group has a relative increase in the percentage of residents staying at home in both categories. However, as indicated by the gaps between the curves, the increase is larger in counties with lower percentages of residents who did not attend high school, which suggests the existence of treatment effect heterogeneity across education.

We formally analyze the treatment effect heterogeneity across education by interacting the percentage of residents who did not attend high school with the dummies of the treatment, days, and states in the triple-difference model. Note that the percentage of residents who did not attend high school is a continuous feature in this analysis. The last column of Table 5 summarizes the results. The coefficient of *OrderImplementation* × *NoHighSchool* is negative and significantly different from zero, suggesting that the orders have a smaller effect on counties with higher percentages of residents who did not attend high school. More specifically, a coefficient of −0.069 means that a 1-percentage point increase in the number of residents who did not attend high school reduces the effect of the orders by 0.069 percentage points (or 0.34%).

The coefficient of *OrderExpiration* × *NoHighSchool* is positive and significantly different from zero, suggesting that the expiration of the orders has a smaller effect on counties with higher percentages of residents who did not attend high school. More specifically, a coefficient of 0.071 means that a 1-percentage point increase in the number of residents who did not attend high school reduces the effect of the expiration of the orders by 0.071 percentage points (or 0.24%). Our estimation shows that the coefficients of *OrderImplementation* × *NoHighSchool* and *OrderExpiration* × *NoHighSchool* have opposite signs, which presents more compelling evidence that the orders have a smaller effect on counties with higher percentages of residents who did not attend high school.

6.1.3. Additional Analyses. We perform similar heterogeneous treatment effect analyses using the SafeGraph's Social Distancing Metrics data. As described in Section 5.3.9, a nice feature of the data is that the devices are aggregated at the census block group instead of county level. These semiindividual-level data provide more direct evidence between those uninsured and less educated and their mobility. More specifically, the data include more than 204,686 census block groups in 3,191 counties.[32] To examine the treatment effect heterogeneity at the census block group level, we use the ACS data that aggregate residents' demographics and social economic status at the census block group level. Because of the large number of census block groups and a long study period, we aggregate the percentage of devices by week instead of day.

Table C4 in Online Appendix C summarizes the results from the alternative data. The number of observations is 204,686 census block groups × 44 weeks = 9,006,184. From column (1), we see that the coefficient of *OrderImplementation* × *Uninsured* is negative and significantly different from zero, which suggests that the orders have a smaller effect on census block groups with higher percentages of uninsured residents. Similarly, from column (2), we see that the coefficient of *OrderImplementation* × *NoHighSchool* is negative and significantly different from zero, which suggests that the orders have a smaller effect on census block groups with higher percentages of residents who did not attend high school. The results from these additional analyses are consistent with those in Sections 6.1.1 and 6.1.2, which provide additional evidence that the orders have heterogeneous effects on those uninsured and less educated.

6.2. Mechanisms of Change

The main purpose of the stay-at-home orders is to reduce person-to-person contacts by limiting the number of nonessential trips made by residents. Our results show that the orders increase the percentage of residents staying at home and therefore, implicitly suggest that the orders reduce the number of trips. In this subsection, we explicitly study the effect of the orders on the number of trips. We differentiate work and nonwork trips because the essentialness of these two types of trips differs. We first quantify the average effect of the orders on the average number of work and nonwork trips per person. We then explore the treatment effect heterogeneity across insurance and education to understand why the orders are less effective for some residents.

6.2.1. Reduction of Work Trips. We estimate the average and heterogeneous effects of the orders on the average number of work trips per person using the difference-in-differences and triple-difference models, respectively.[33] Table 6 summarizes the results. From the first column, we see that the coefficient of *OrderImplementation* is negative and significantly different from zero, suggesting that the orders reduce the average number of work trips per person. More specifically, a coefficient of −0.053 means that the orders reduce the average number of work trips by 0.053 (or 7.87%).

From the second column, we see that the coefficient of *OrderImplementation* × *Uninsured* is positive and significantly different from zero, suggesting that the reduction in the number of work trips correlates with the percentage of county residents without health insurance.[34] Finally, from the third column, we see that the coefficient of *OrderImplementation* × *NoHighSchool* is positive, suggesting that the reduction in the number of work trips correlates with the percentage of county residents who did not attend high school. Our results are consistent with those reported by media outlets—vulnerable populations are disproportionately affected by COVID-19 because many of them work for essential businesses that put them at higher risks.[35]

6.2.2. Reduction of Nonwork Trips. We now estimate the average and heterogeneous effects of the orders on the average number of nonwork trips per person using the difference-in-differences and triple-difference models, respectively.[36] Table 7 summarizes the results. From the first column, we see that the coefficient of *OrderImplementation* is negative and significantly different from zero, suggesting that the orders reduce the average number of nonwork trips per person. More specifically, a coefficient of −0.183 means that the orders reduce the average number of nonwork trips by 0.183 (or 6.50%).

From the second column, we see that the coefficient of *OrderImplementation* × *Uninsured* is not significantly different from zero at the 10% significance level, suggesting that the reduction in the number of nonwork trips does not correlate with the percentage of county residents without health insurance. Similarly, the insignificance of the coefficient of *OrderImplementation* × *NoHighSchool* in the third column suggests that the

Table 6. Average and Heterogeneous Effects of the Orders on Work Trips

	(1) All		(2) *Uninsured*		(3) *NoHighSchool*	
Variable	Coefficient	Standard error	Coefficient	Standard error	Coefficient	Standard error
OrderImplementation	−0.053***	0.014	−0.117***	0.027	−0.092***	0.003
× *TargetFeature*			0.006***	0.002	0.002***	0.000
OrderExpiration	Included		Included		Included	
× *TargetFeature*			Included		Included	
CountyFeatures	Included		Included		Included	
DayFixedEffects	Included		Included		Included	
× *TargetFeature*			Included		Included	
StateFixedEffects	Included		Included		Included	
× *TargetFeature*			Included		Included	
Number of observations	943,975		943,975		943,975	
Adjusted R^2	0.390		0.411		0.412	

Notes. Column (1) summarizes the result from average effect analysis. Columns (2) and (3) summarize the results from heterogeneous effect analyses, where the target features are *Uninsured* and *NoHighSchool*, respectively. Robust standard errors are clustered by state.
$^{***}p < 0.01$.

Table 7. Average and Heterogeneous Effects of the Orders on Nonwork Trips

	(1) All		(2) *Uninsured*		(3) *NoHighSchool*	
Variable	Coefficient	Standard error	Coefficient	Standard error	Coefficient	Standard error
OrderImplementation	−0.183***	0.036	−0.188***	0.040	−0.198***	0.040
× *TargetFeature*			0.003	0.003	0.003	0.003
OrderExpiration	Included		Included		Included	
× *TargetFeature*			Included		Included	
CountyFeatures	Included		Included		Included	
DayFixedEffects	Included		Included		Included	
× *TargetFeature*			Included		Included	
StateFixedEffects	Included		Included		Included	
× *TargetFeature*			Included		Included	
Number of observations	943,975		943,975		943,975	
Adjusted R^2	0.451		0.466		0.468	

Notes. Column (1) summarizes the result from average effect analysis. Columns (2) and (3) summarize the results from heterogeneous effect analyses, where the target features are *Uninsured* and *NoHighSchool*, respectively. Robust standard errors are clustered by state.
***$p < 0.01$.

reduction in the number of nonwork trips does not correlate with the percentage of county residents who did not attend high school. Our results do not support speculations that vulnerable populations are less aware of the COVID-19 pandemic and more likely to leave their homes for unnecessary trips.

Comparing Tables 6 and 7, we see that, on average, the orders are more effective in reducing the number of nonwork trips than work trips (see the first columns). From the last two columns, we see that the percentage of county residents without health insurance or who did not attend high school correlates with the reduction in the number of work trips but does not correlate with the reduction in the number of nonwork trips.

These results suggest that residents without health insurance or who did not attend high school are less likely to stay at home because they cannot reduce the number of work trips as much as insured and well-educated residents. The reason, in part, is that uninsured and less educated residents are more likely to work in businesses that do not allow or have the means for employees to work from home.[37]

7. Conclusion

COVID-19 has swept the entire United States from coast to coast and from north to south, infecting millions of Americans. The pandemic has also ignited hot debates regarding the effectiveness of the stay-at-home orders and sociodemographic disparities. Given that some residents have already voluntarily stayed at home, whether the orders are still necessary and how much they can help increase the percentage of residents staying at home are unclear. The alarming disparities exacerbated by COVID-19 have shocked many policy makers, healthcare providers, and researchers. Although some believe that these disparities are because of underlying health conditions across different populations, others point out the sociodemographic inequalities that have existed for a long time in society.

This study contributes to these debates by answering several important questions. First, we find that the orders are effective in increasing the percentage of residents staying at home. More specifically, we use a difference-in-differences model to compare the counties that implemented the orders with those that did not before and after the implementation of the orders. We find that the orders increase the number of residents staying at home by 2.832 percentage points (or 11.25%). Our result is robust to alternative model specifications, treatment groups, and dependent variables.

Second, we find that the effect of the orders is heterogeneous across counties with different percentages of vulnerable populations. Leveraging the variations across counties in residents' demographics, social, and economic status, we use a triple-difference model to analyze the treatment effect heterogeneity across insurance and education. We find that the orders have a smaller effect on counties with higher percentages of residents without insurance or who did not attend high school.

Finally, to understand the mechanisms behind our results, we analyze the effect of the orders on reducing the number of work and nonwork trips. We find that the orders reduce the average number of work trips per person by 0.053 (or 7.87%) and the average number of nonwork trips per person by 0.183 (or 6.50%). We also find significant treatment effect heterogeneity across counties. More specifically, the orders are less effective in reducing the number of work trips in counties with higher percentages of uninsured residents and residents who did not attend high school.

This study supports the implementation of the stay-at-home orders as a measure to contain the spread of the pandemic (see Online Appendix E for the impact of mobility on infections). As our results indicate, the United States would be in a much worse situation if none of the states implemented the orders. This study also highlights important sociodemographic disparities exacerbated by the pandemic and stay-at-home orders. Our results suggest that many uninsured and less educated residents could not stay at home because their jobs require them to be physically present in companies or with customers, which increases the risk of infection. Policy makers must take into account the differences in residents' socioeconomic status when developing new policies or allocating healthcare resources.

Endnotes

[1] See https://coronavirus.jhu.edu/map.html.

[2] See https://www.businessinsider.com/coronavirus-lockdown-united-states-entire-country-anthony-fauci-cdc-2020-4.

[3] See https://www.nytimes.com/interactive/2020/us/coronavirus-stay-at-home-order.html.

[4] See https://www.kff.org/coronavirus-policy-watch/stay-at-home-orders-to-fight-covid19.

[5] See https://www.oregonlive.com/coronavirus/2020/03/whats-the-penalty-for-breaking-oregons-new-stay-home-order.html.

[6] See https://www.multistate.us/pages/covid-19-policy-tracker.

[7] See https://www.ncbi.nlm.nih.gov/pubmed/32191675.

[8] See https://www.fema.gov/news-release/2020/03/13/covid-19-emergency-declaration.

[9] See https://www.kff.org/health-costs/issue-brief/state-data-and-policy-actions-to-address-coronavirus.

[10] See https://www.cdc.gov/coronavirus/2019-ncov/prevent-getting-sick/social-distancing.html.

[11] See https://www.nytimes.com/interactive/2020/04/02/us/coronavirus-social-distancing.html.

[12] See https://www.nbcnews.com/politics/politics-news/hundreds-gather-north-carolina-missouri-protest-stay-home-orders-n1188936.

[13] See https://www.theguardian.com/us-news/2017/feb/16/homeless-count-population-america-shelters-people.

[14] See https://www.bbc.com/future/article/20200420-coronavirus-why-some-racial-groups-are-more-vulnerable.

[15] See https://www.cdc.gov/mmwr/volumes/69/wr/mm6915e3.htm.

[16] See https://www.cdc.gov/mmwr/volumes/69/wr/mm6915e3.htm.

[17] See https://www.texastribune.org/2020/03/10/coronavirus-texas-poses-challenges-people-without-insurance/.

[18] See https://www.bbc.com/future/article/20200420-coronavirus-why-some-racial-groups-are-more-vulnerable.

[19] See https://www.nytimes.com/interactive/2020/us/coronavirus-stay-at-home-order.html.

[20] See https://data.covid.umd.edu/.

[21] See https://www.census.gov/programs-surveys/acs/data.html.

[22] See https://www.okc.gov/Home/Components/News/News/3321/18 and http://edmondok.com/1568/COVID-19-Coronavirus-Information.

[23] Including more features may introduce multicollinearity issues and does not change the main conclusion of this study.

[24] An alternative approach is to categorize counties into groups based on the values of the target feature. For example, we can categorize counties into two groups based the median of the target feature: $HighTarget_i = \mathbb{1}\{Target_i \geq Median\}$ and $LowTarget_i = \mathbb{1}\{Target_i < Median\}$. This approach does not change the main conclusion of this study.

[25] That is, the mean is an unweighted average of county outcomes/features.

[26] We conducted a probit analysis at the state level (see Table B1 in Online Appendix B) and did not find the implementation of stay-at-home orders to be correlated with race, ethnicity, education, insurance, or pretreatment outcome.

[27] See https://www.bu.edu/sph/news/articles/2020/tracking-covid-19-policies/.

[28] Using alternative control (e.g., state that implemented the orders before March 25, 2020) and treatment groups (e.g., states that implemented the orders after March 30, 2020) does not change the main conclusion of this robustness check.

[29] Using an alternative period as a baseline does not change the main conclusion of this robustness check.

[30] See https://www.safegraph.com/.

[31] Because our data include a large number of census block groups (more than 200,000), we aggregate the percentage of devices by week instead of day.

[32] To protect individuals' privacy, SafeGraph does not include census block groups with fewer than five observable devices.

[33] In Online Appendix D, the left panel of Figure D1 compares the number of work trips in the control and treatment groups over different days. Figures D2 and D3 in Online Appendix D compare the number of work trips by percentage of residents without health insurance or high school diploma.

[34] The Kaiser Family Foundation estimated that one in five low-wage workers lacked health coverage in 2018. Some of these workers may in fact be eligible for Medicaid or Affordable Care Act marketplace coverage but did not enroll. More details can be found at https://www.kff.org/coronavirus-covid-19/issue-brief/double-jeopardy-low-wage-workers-at-risk-for-health-and-financial-implications-of-covid-19/.

[35] Rho et al. (2020) studied the basic demographic profile of workers in frontline industries and found that nearly 3 in 10 workers in the building cleaning services industry are uninsured.

[36] In Online Appendix D, the right panel of Figure D1 compares the number of nonwork trips in the control and treatment groups over different days. Figures D4 and D5 in Online Appendix D compare the number of nonwork trips by percentage of residents without health insurance or high school diploma.

[37] See https://time.com/5800930/how-coronavirus-will-hurt-the-poor/.

References

Abouk R, Heydari B (2021) The immediate effect of COVID-19 policies on social-distancing behavior in the United States. *Public Health Rep.* 136(2):245–252.

Alexander D, Karger E (2020) Do stay-at-home orders cause people to stay at home? Effects of stay-at-home orders on consumer behavior. Working paper, Federal Reserve Bank of Chicago, Chicago.

Andersen M (2020) Early evidence on social distancing in response to COVID-19 in the United States. Working paper, Department of Economics, University of North Carolina at Greensboro, Greensboro, NC.

Angrist JD, Pischke JS (2008) *Mostly Harmless Econometrics: An Empiricist's Companion* (Princeton University Press, Princeton, NJ).

Askitas N, Tatsiramos K, Verheyden B (2020) Lockdown strategies, mobility patterns and COVID-19. Working paper, IZA Institute of Labor Economics, Bonn, Germany.

Bapna R, Ramaprasad J, Umyarov A (2018) Monetizing freemium communities: Does paying for premium increase social engagement? *Management Inform. Systems Quart.* 42(3):719–735.

Barrios JM, Hochberg Y (2020) Risk perception through the lens of politics in the time of the COVID-19 pandemic. Working paper, Booth School of Business, University of Chicago, Chicago.

Bonaccorsi G, Pierri F, Cinelli M, Flori A, Galeazzi A, Porcelli F, Schmidt AL, , et al. (2020) Economic and social consequences of human mobility restrictions under COVID-19. *Proc. Natl. Acad. Sci. USA* 117(27):15530–15535.

Brodeur A, Grigoryeva I, Kattan L (2021) Stay-at-home orders, social distancing, and trust. *J. Population Econom.*, ePub ahead of print June 19, https://doi.org/10.1007/s00148-021-00848-z.

Brot-Goldberg ZC, Chandra A, Handel BR, Kolstad JT (2017) What does a deductible do? The impact of cost-sharing on healthcare prices, quantities, and spending dynamics. *Quart. J. Econom.* 132(3):1261–1318.

Bryant P, Elofsson A (2020) Estimating the impact of mobility patterns on COVID-19 infection rates in 11 European countries. *PeerJ* 8(2020):e9879.

Brzezinski A, Kecht V, Van Dijcke D (2020a) The cost of staying open: Voluntary social distancing and lockdowns in the U.S. Working paper, Department of Economics, University of Oxford, Oxford, United Kingdom.

Brzezinski A, Deiana G, Kecht V, Van Dijcke D (2020b) The COVID-19 pandemic: Government vs. community action across the United States. INET Oxford Working Paper No. 2020-06, Institute for New Economic Thinking at the Oxford Martin School, University of Oxford, Oxford, United Kingdom.

Brzezinski A, Kecht V, Van Dijcke D, Wright AL (2020c) Belief in science influences physical distancing in response to COVID-19 lockdown policies. Working paper, Becker Friedman Institute for Economics, University of Chicago, Chicago.

Bushman K, Pelechrinis K, Labrinidis A (2020) Effectiveness and compliance to social distancing during COVID-19. Working paper, School of Computing and Information, University of Pittsburgh, Pittsburgh.

Charoenwong B, Kwan A, Pursiainen V (2020) Social connections with COVID-19-affected areas increase compliance with mobility restrictions. *Sci. Adv.* 6(47):eabc3054.

Chinazzi M, Davis JT, Ajelli M, Gioannini C, Litvinova M, Merler S, Pastore Y Piontto A, et al. (2020) The effect of travel restrictions on the spread of the 2019 novel coronavirus (COVID-19) outbreak. *Science* 368(6489):395–400.

Chiou L, Tucker C (2020) Social distancing, internet access and inequality. Working paper, Department of Economics, Occidental College, Los Angeles.

Cook J, Newberger N, Smalling S (2020) The spread of social distancing. *Econom. Lett.* 196(2020):109511.

Cronin CJ, Evans WN (2020) Private precaution and public restrictions: What drives social distancing and industry foot traffic in the COVID-19 era? Working paper, Department of Economics, University of Notre Dame, Notre Dame, IN.

Dasgupta N, Jonsson Funk M, Lazard A, White BE, Marshall SW (2020) Quantifying the social distancing privilege gap: A longitudinal study of smartphone movement. Working paper, Injury Prevention Research Center, University of North Carolina at Chapel Hill, Chapel Hill, NC.

Dave D, Friedson AI, Matsuzawa K, Sabia JJ (2021) When do shelter-in-place orders fight COVID-19 best? Policy heterogeneity across states and adoption time. *Econom. Inquiry* 59(1):29–52.

Ding W, Levine R, Lin C, Xie W (2020) Social distancing and social capital: Why U.S. counties respond differently to COVID-19. Working paper, Faculty of Business and Economics, University of Hong Kong, Hong Kong.

Engle S, Stromme J, Zhou A (2020) Staying at home: Mobility effects of COVID-19. Working paper, Department of Economics, University of Wisconsin–Madison, Madison, WI.

Fang H, Wang L, Yang Y (2020) Human mobility restrictions and the spread of the novel coronavirus (2019-nCoV) in China. *J. Public Econom.* 191(2020):104272.

Fowler JH, Hill SJ, Levin R, Obradovich N (2021) Stay-at-home orders associate with subsequent decreases in COVID-19 cases and fatalities in the United States. *PLoS One* 16(6):e0248849.

Gao S, Rao J, Kang Y, Liang Y, Kruse J, Doepfer D, Sethi AK, Reyes JRM, Yandell BS, Patz JA (2020) Mobile phone location data reveal the effect and geographic variation of social distancing on the spread of the COVID-19 epidemic. *JAMA Network Open* 3(9): e2020485.

Gatto M, Bertuzzo E, Mari L, Miccoli S, Carraro L, Casagrandi R, Rinaldo A (2020) Spread and dynamics of the COVID-19 epidemic in Italy: Effects of emergency containment measures. *Proc. Natl. Acad. Sci. USA* 117(19):10484–10491.

Glaeser EL, Gorback C, Redding SJ (2020) JUE insight: How much does COVID-19 increase with mobility? Evidence from New York and four other U.S. cities. *J. Urban Econom.*, ePub ahead of print October 21, https://doi.org/10.1016/j.jue.2020.103292.

Grossman G, Kim S, Rexer JM, Thirumurthy H (2020) Political partisanship influences behavioral responses to governors' recommendations for COVID-19 prevention in the United States. *Proc. Natl. Acad. Sci. USA* 117(39):24144–24153.

Guo J (2020) The impact of statewide stay-at-home orders: Estimating the heterogeneous effects using GPS data from mobile devices. Working paper, Center for Research on the Wisconsin Economy, University of Wisconsin–Madison, Madison, WI.

Guo ZD, Wang ZY, Zhang SF, Li X, Li L, Li C, Cui Y, et al. (2020) Aerosol and surface distribution of severe acute respiratory syndrome coronavirus 2 in hospital wards, Wuhan, China, 2020. *Emerging Infectious Diseases* 26(7):1583–1591.

Gupta S, Nguyen TD, Rojas FL, Raman S, Lee B, Bento A, Simon KI, Wing C (2020) Tracking public and private responses to the COVID-19 epidemic: Evidence from state and local government actions. NBER Working Paper No. 27027, National Bureau of Economic Research, Cambridge, MA.

Huang D (2020) How effective is social distancing? Working paper, Monash Business School, Monash University, Melbourne, VIC, Australia.

Jacobsen GD, Jacobsen KH (2020) Statewide COVID-19 stay-at-home orders and population mobility in the United States. *World Medicine Health Policy*, ePub ahead of print July 29.

Jay J, Bor J, Nsoesie E, Lipson SK, Jones DK, Galea S, Raifman J (2020) Neighbourhood income and physical distancing during the COVID-19 pandemic in the United States. *Nature Human Behav.* 4(12):1294–1302.

Klein B, LaRock T, McCabe S, Torres L, Privitera F, Lake B, Kraemer MUG, et al. (2020) Assessing changes in commuting and individual mobility in major metropolitan areas in the United States during the COVID-19 outbreak. Working paper, Network Science Institute, Northeastern University, Boston.

Kraemer MUG, Yang C-H, Gutierrez B, Wu C-H, Klein B, Pigott DM, du Plessis L, et al. (2020) The effect of human mobility and control measures on the COVID-19 epidemic in China. *Science* 368(6490):493–497.

Li Q, Guan X, Wu P, Wang X, Zhou L, Tong Y, Ren R, et al. (2020) Early transmission dynamics in Wuhan, China, of novel coronavirus-infected pneumonia. *New England J. Medicine* 382(13):1199–1207.

Lin Z, Meissner CM (2020) Health vs. wealth? Public health policies and the economy during COVID-19. Working paper, California Center for Population Research, University of California at Los Angeles, Los Angeles.

Luther WJ (2020) Behavioral and policy responses to COVID-19: Evidence from Google mobility data on state-level stay-at-home orders. Working paper, American Institute for Economic Research, Florida Atlantic University, Boca Raton, FL.

Lyu W, Wehby GL (2020) Shelter-in-place orders reduced COVID-19 mortality and reduced the rate of growth in hospitalizations. *Health Affairs* 39(9):1615–1623.

McGrail DJ, Dai J, McAndrews KM, Kalluri R (2020) Enacting national social distancing policies corresponds with dramatic reduction in COVID-19 infection rates. *PLoS One* 15(7):e0236619.

Painter M, Qiu T (2021) Political beliefs affect compliance with government mandates. *J. Econom. Behav. Organ.* 185(2021):688–701.

Rho HJ, Brown H, Fremstad S (2020) A basic demographic profile of workers in frontline industries. Center for Economic and Policy Research. Accessed January 25, 2021, https://cepr.net/a-basic-demographic-profile-of-workers-in-frontline-industries/.

Silva WTAF (2020) Per capita death and infection rates should be avoided in international comparisons. *Public Health* 186(2020): 18–19.

Sun S, Lu SF, Rui H (2020) Does telemedicine reduce emergency room congestion? Evidence from New York State. *Inform. Systems Res.* 31(3):972–986.

van Doremalen N, Bushmaker T, Morris DH, Holbrook MG, Gamble A, Williamson BN, Tamin A, et al. (2020) Aerosol and surface stability of SARS-CoV-2 as compared with SARS-CoV-1. *New England J. Medicine* 382(16):1564–1567.

Wang G (2020) Stay at home to stay safe: Effectiveness of stay-at-home orders in containing the COVID-19 pandemic. Working paper, Jindal School of Management, University of Texas at Dallas, Richardson, TX.

Weill JA, Stigler M, Deschenes O, Springborn MR (2020) Social distancing responses to COVID-19 emergency declarations strongly differentiated by income. *Proc. Natl. Acad. Sci. USA* 117(33): 19658–19660.

Wellenius GA, Vispute S, Espinosa V, Fabrikant A, Tsai TC, Hennessy J, Dai A, et al. (2021) Impacts of social distancing policies on mobility and COVID-19 case growth in the US. *Nature Comm.* 12(1):1–7.

Wright AL, Sonin K, Driscoll J, Wilson J (2020) Poverty and economic dislocation reduce compliance with COVID-19 shelter-in-place protocols. *J. Econom. Behav. Organ.* 180(2020):544–554.

Yang W, Shaff J, Shaman J (2021) Effectiveness of non-pharmaceutical interventions to contain COVID-19: A case study of the 2020 spring pandemic wave in New York City. *J. Royal Soc. Interface* 18(175):20200822.

Yilmazkuday H (2021) Stay-at-home works to fight against COVID-19: International evidence from Google mobility data. *J. Human Behav. Soc. Environ.* 31(1-4):210–220.